

University of Prishtina “Hasan PRISHTINA”



Faculty of Electrical and Computer Engineering
Master in Computer and Software Engineering, 2025-2026
Course: Data Preparation and Visualization

Week XIII
Data Visualization I

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Metrics for Performance Evaluation

➤ Today we are going to learn about:

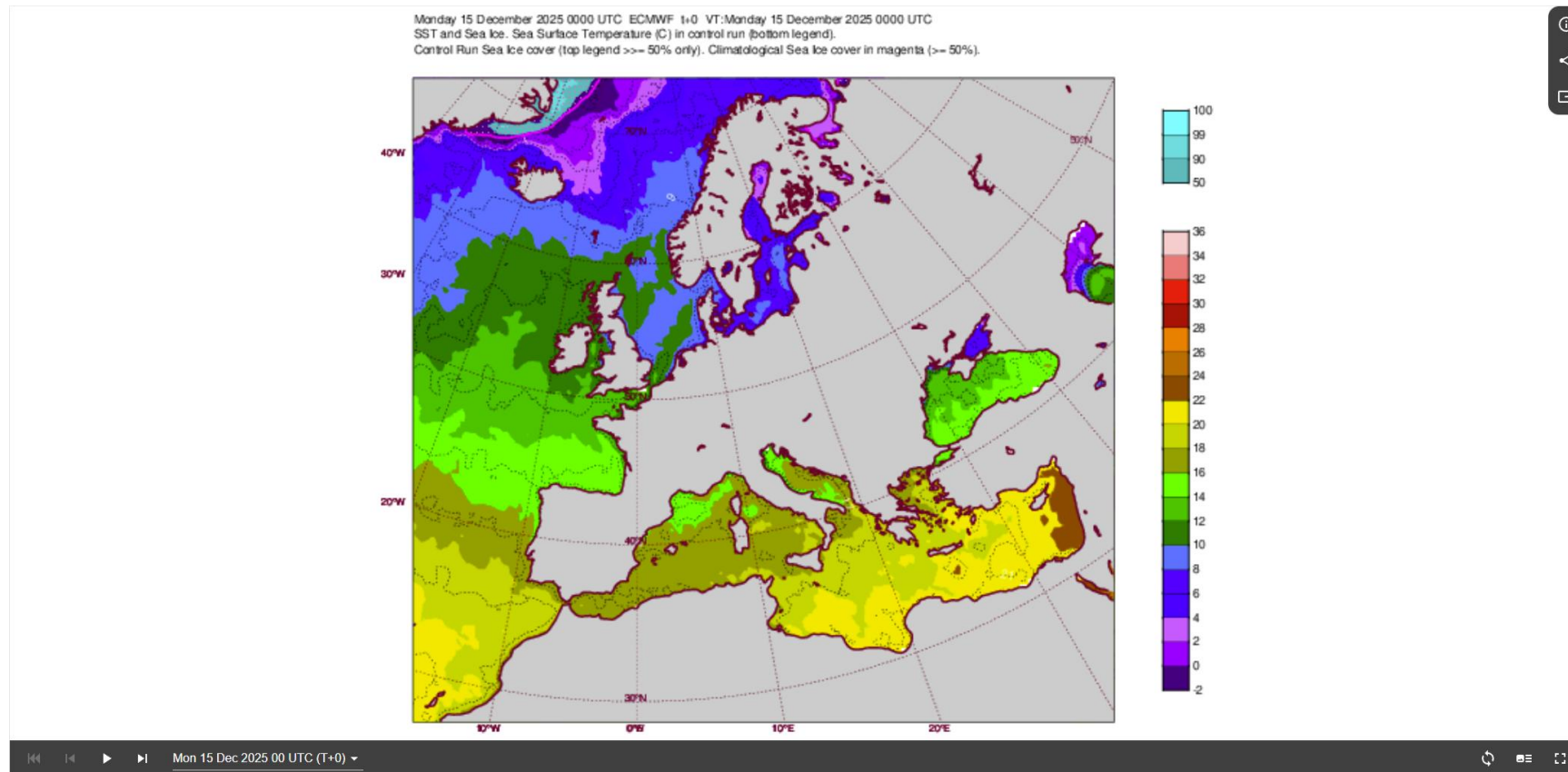
- Data visualization

Visualization

- Visualization is the conversion of data into a visual or tabular format so that the characteristics of the data and the relationships among data items or attributes can be analysed or reported;
- Visualization of data is one of the **most powerful** and appealing techniques for **data exploration**:
 - Humans have a well-developed ability to analyze large amounts of information that is presented visually;
 - Can detect general patterns and trends;
 - Can detect outliers and unusual patterns.

Example: Sea Surface Temperature

- The following shows the Sea Surface Temperature (SST) for Europe, 15.12.2025:
 - Tens of thousands of data points are summarized in a single figure:



Representation

- Is the mapping of information to a visual format;
- **Data objects**, their **attributes**, and the **relationships** among **data objects** are translated into **graphical elements** such as **points**, **lines**, **shapes**, and **colours**.
- **Example:**
 - Objects are often represented as points;
 - Their attribute values can be represented as the **position of the points** or the **characteristics of the points**, e.g., colour, size, and shape;
 - If position is used, then the relationships of points, i.e., whether they form groups or a point is an outlier, is easily perceived.

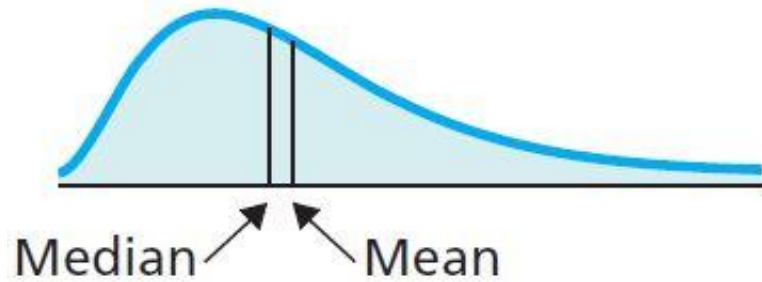
Arrangement

- Is the placement of visual elements within a display;
- Can make a large difference in how easy it is to understand the data;
- Example:

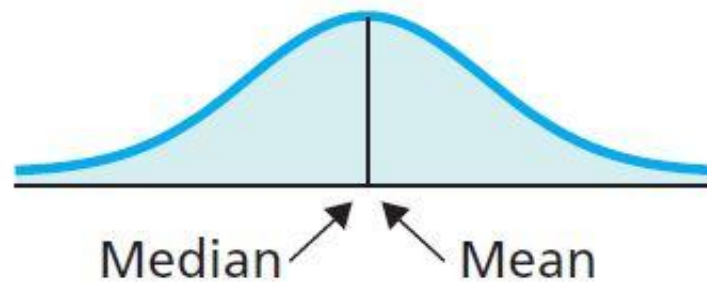
	1	2	3	4	5	6
1	0	1	0	1	1	0
2	1	0	1	0	0	1
3	0	1	0	1	1	0
4	1	0	1	0	0	1
5	0	1	0	1	1	0
6	1	0	1	0	0	1
7	0	1	0	1	1	0
8	1	0	1	0	0	1
9	0	1	0	1	1	0

	6	1	3	2	5	4
4	1	1	1	0	0	0
2	1	1	1	0	0	0
6	1	1	1	0	0	0
8	1	1	1	0	0	0
5	0	0	0	1	1	1
3	0	0	0	1	1	1
9	0	0	0	1	1	1
1	0	0	0	1	1	1
7	0	0	0	1	1	1

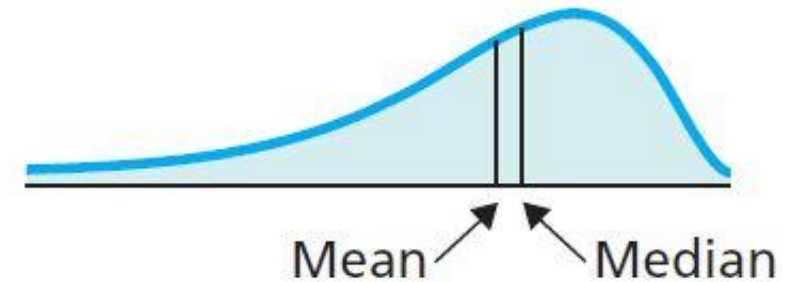
Data Distribution Shapes



(a) Right skewed



(b) Symmetric



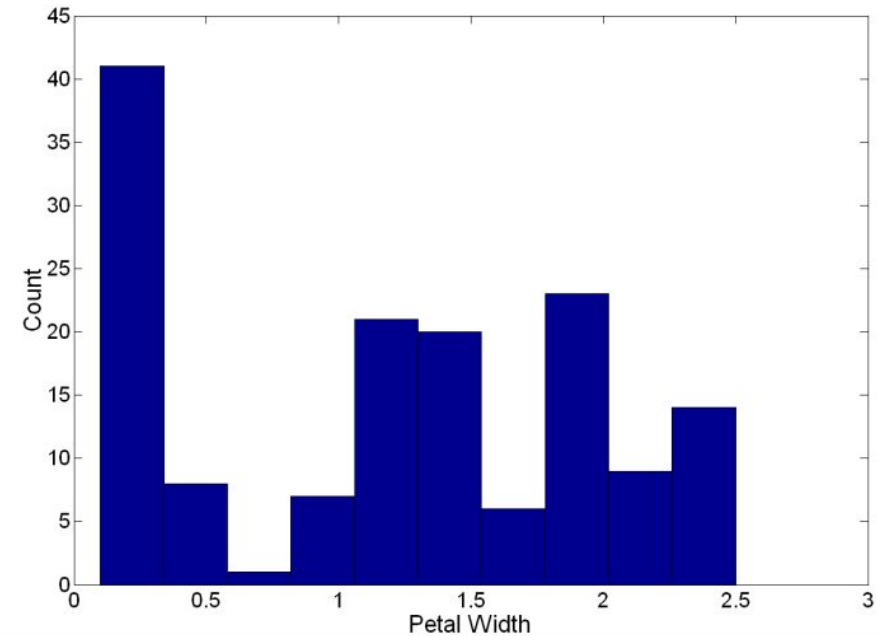
(c) Left skewed

Visualization Techniques: Histograms

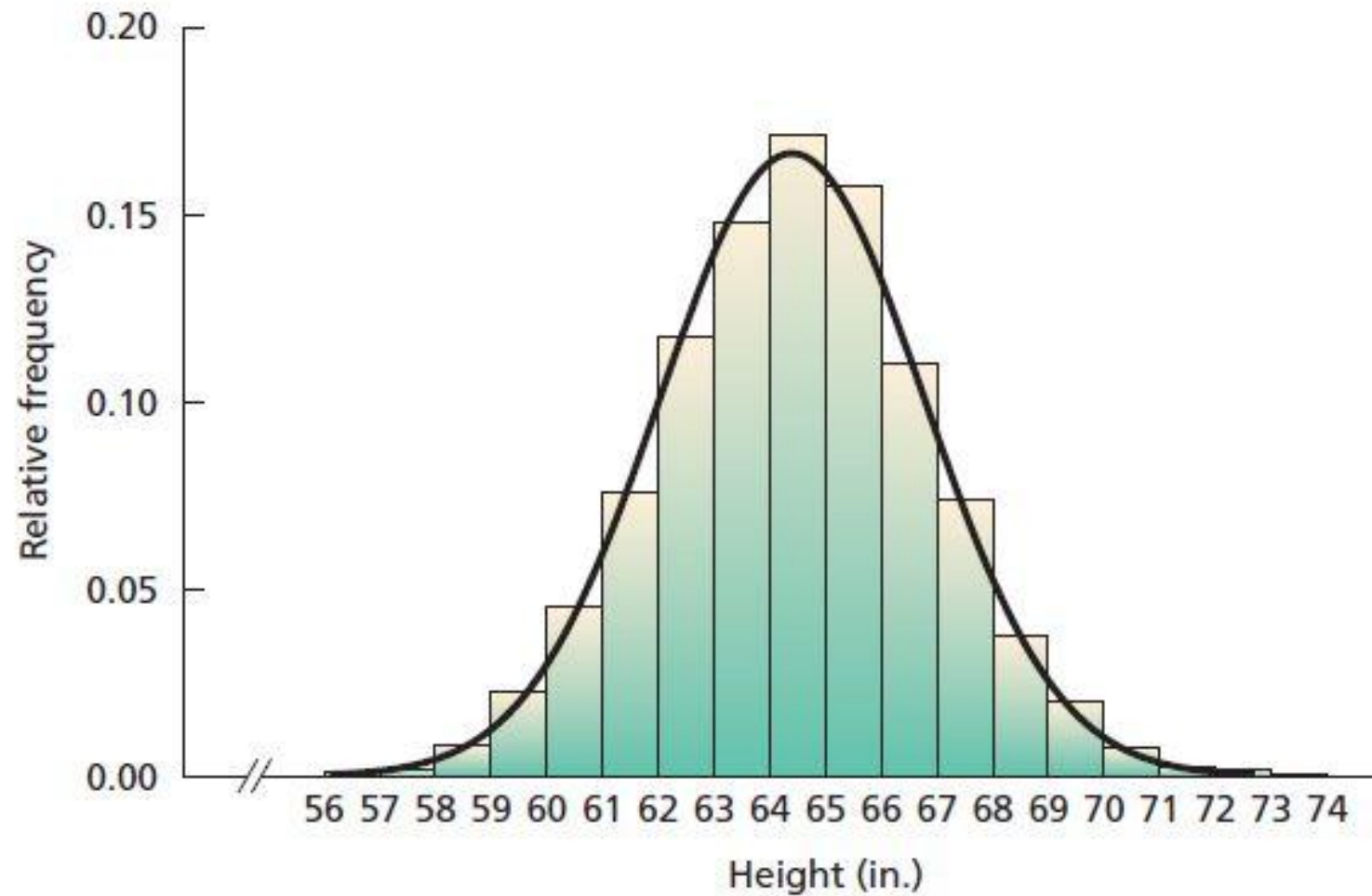
➤ Histogram:

- Usually shows the distribution of values of a single variable;
- Divide the values into bins and show a bar plot of the number of objects in each bin;
- The height of each bar indicates the number of objects;
- Shape of histogram depends on the number of bins.

➤ Example: Petal Width

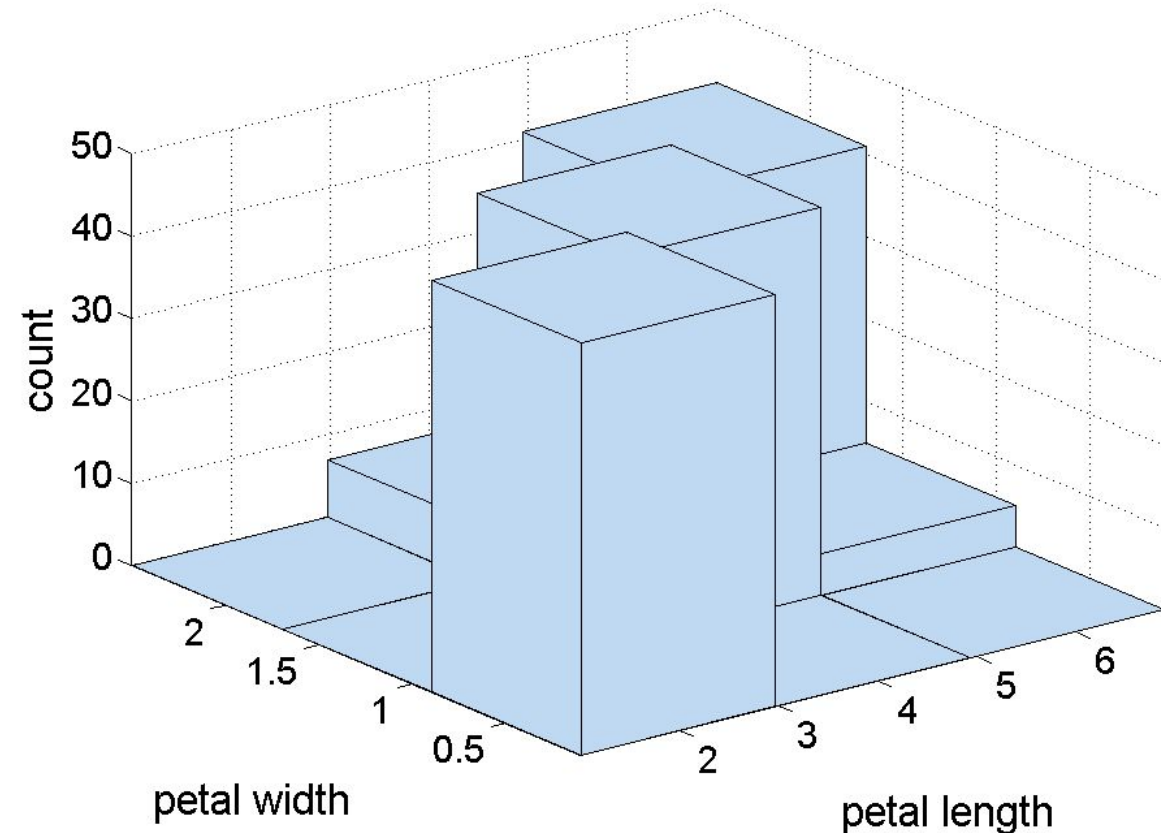


Visualization Techniques: Histograms



Two - Dimensional Histograms

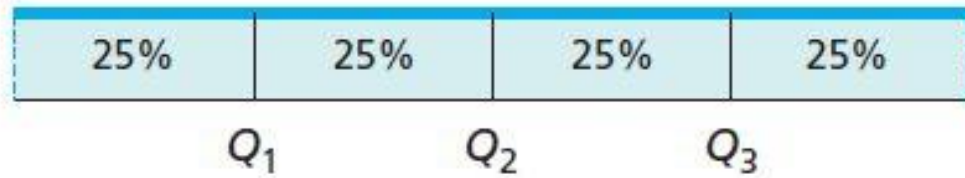
- Show the joint distribution of the values of two attributes;
- Example: petal width and petal length:
 - What does this tell us?



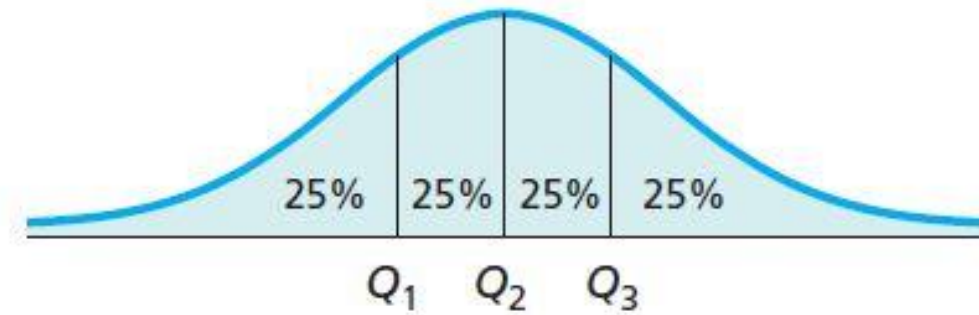
Visualization Techniques: Quartiles

- Percentile: divides the data into hundredths (100 equal parts) $P_1, P_2, \dots, P_{99}$;
- Deciles: divides the data into tenths (10 equal parts);
- Quintiles: divides the data into fifths (5 equal parts);
- Quartiles: divides the data into quarters (4 equal parts) Q_1, Q_2, Q_3 .

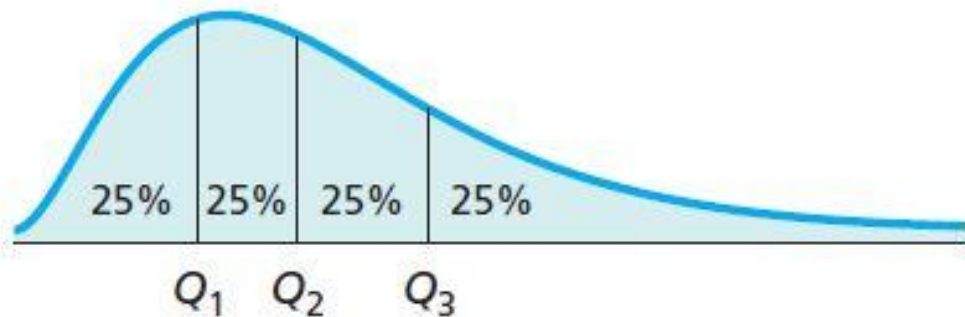
Visualization Techniques: Quartiles



(a) Uniform



(b) Bell shaped



(c) Right skewed



(d) Left skewed

Five Number Summary

➤ **Five-Number Summary:** min, Q_1 , Q_2 , Q_3 , Max;

➤ **Interquartile range (IQR):**

$$\text{IQR} = Q_3 - Q_1$$

➤ **Limits of the dataset:**

- Lower limit = $Q_1 - 1.5 \times \text{IQR}$
- Upper limit = $Q_3 + 1.5 \times \text{IQR}$

➤ **Outliers:**

- The objects below the lower limit and above the upper limit are potential outliers.

Five Number Summary

➤ Find the 5 Number Summary of the following numbers:

3 12 7 40 9 14 18 15 17

Step 1: Sort the numbers from lowest to highest

3 7 9 12 14 15 17 18 40

Step 2: Identify the Median

3 7 9 12 **14** 15 17 18 40

Step 3: Identify the Smallest and Largest numbers

3 7 9 12 **14** 15 17 18 **40**

Step 4: Identify the Median between the smallest number and the Median for the entire set of data, and between that Median and the largest number in the set.

3 7 **9** 12 **14** 15 **17** 18 **40**

Five Number Summary

➤ These are the five numbers in the 5 Number Summary:

3 7 9 12 14 15 17 18 40

3 7 9 12 14 15 17 18 40

Smallest number in the set:

3

Median between the smallest number and the median:

9

Median of the entire set:

14

Median between the largest number and the median:

17

Largest number in the set:

40

Five Number Summary – Class example

➤ Find the 5 Number Summary of the following numbers:

5 11 2 28 35 17 21 54 18

- ____ - Smallest number in the set
- ____ - Median between the smallest number and the median
- ____ - Median of the entire set
- ____ - Median between the largest number and the median
- ____ - Largest number in the set

Questions?

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